

DeITor Car Inc. – Car Rental System

Overview

DeITor Car Inc. is a Python-based CLI application for managing a car rental business. It demonstrates object-oriented design, design patterns (Factory Method, Repository, Singleton, Strategy), and SQLite database integration.

The system supports customer and admin roles, car management, bookings, and rental approvals/returns.

Installation & Configuration

Prerequisites

- Windows 10/11 (tested environment)
- Python 3.10+ (for development)
- SQLite (bundled with Python)
- PyInstaller if building executables

Steps

1. Clone or download this repository.
2. `git clone <your_repo_url>`
3. `cd DeITorCarInc_Full`
4. (Optional, for development) Create and activate a virtual environment:
5. `python -m venv venv`
6. `venv\Scripts\activate`
7. `pip install -r requirements.txt`
8. Run the CLI app:
9. `python run.py`

Build Executable

```
pyinstaller run.py --onefile --name DeITorCarInc
```

The built executable will be in `dist/DeITorCarInc.exe`.

How to Use

- Run DelTorCarInc.exe (or python run.py).
- At the login menu, you can:
 - Customer: Register, view cars, make bookings, return rentals.
 - Admin: Manage cars, users, and approve/reject rentals.

Default seeded accounts:

- Admin: admin@example.com / 0000
- Customer: alice@example.com / 1111

Project Structure

Your del_tor_cars package is modular:

del_tor_cars/

__init__.py	→ makes package importable
app.py	→ CLI application (menus, navigation, orchestration)
config.py	→ configuration constants (DB name, date formats)
db.py	→ Database connection, schema initialization, seeding
errors.py	→ custom exception classes
factories.py	→ Factory Method implementations (RepoFactory, ServiceFactory)
hashing.py	→ hashing strategies (SHA-256, future bcrypt/argon2)
ifaces.py	→ abstract interfaces for factories
models.py	→ domain models (User, Car)
pricing.py	→ pricing strategies (standard, dynamic)
utils.py	→ utility helpers (dates, password hashing)
validators.py	→ input validation helpers

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└─ repos/ → Repositories (UserRepo, CarRepo, RentalRepo)

└─ services/ → Business services (AuthService, BookingService, AdminService)

Licensing

MIT License

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Known Issues

- Passwords currently hashed with SHA-256 (consider upgrading to bcrypt/argon2).
 - CLI only — no web or mobile front-end yet.
 - Error messages are text-only; no localization.
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Credits

- Developer: Mark Dela Torre
- Built as part of a Professional Software Engineering project at Yoobee.